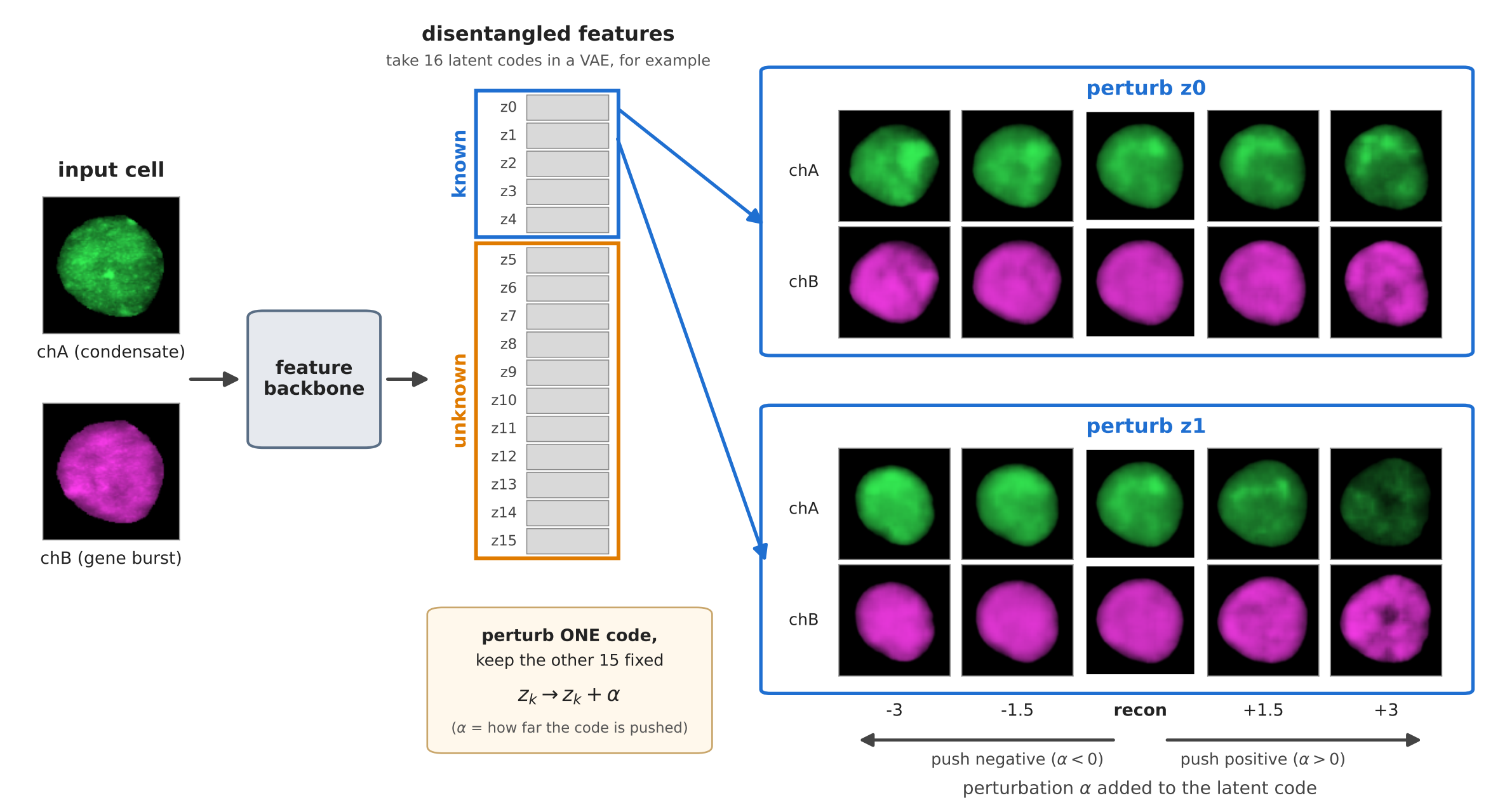

disentangled features
take 16 latent codes in a VAE, for example
input cell
chA (condensate)
chB (gene burst)
feature backbone
known
z0
z1
z2
z3
z4
unknown
z5
z6
z7
z8
z9
z10
z11
z12
z13
z14
z15
perturb ONE code,
keep the other 15 fixed
$z_k \rightarrow z_k + \alpha$
(α = how far the code is pushed)
perturb z0
chA
chB
perturb z1
chA
chB
-3
-1.5
recon
+1.5
+3
push negative ($\alpha < 0$)
push positive ($\alpha > 0$)
perturbation α added to the latent code